Polymer Promises

A look at the research in the field of plastic electronics

Nimish Sawant
nimish.sawant@9dot9.in

Whenever we talk about clean and green energy alternatives, plastic is the last thing that will come to anyone's mind. Well, how would you feel if we told you that a team of researchers has created the first completely plastic Solar Cell?

A subset of electronics

We will get to the plastic solar cell in a bit. But first, welcome to the field of plastic electronics – a branch of organic electronics that uses organic carbon-based polymers which are semi-conductive in nature to build electronics instead of silicon. You are familiar with organic polymers, which are all around us in the form of plastic bags, bottles to even solar panels.

Plastic electronics basically comprise a plastic substrate made from organic material onto which electronic circuits are printed in the form of flattened sheets. Unlike silicon-based electronics, plastic electronics, by their very attributes are much more robust and can easily be flexed or bent while maintaining the conductivity at the same time. Inorganic materials in electronics are brittle in nature and thereby cannot be subjected to flexing without affecting the overall conductivity in some way or the other.

Ongoing research

Researchers at Princeton University led by Yueh Lin Loo have developed plastics that can conduct electricity. It was initially difficult to produce plastic which excelled in three attributes, namely translucence (ability to be semi-transparent), malleability (ability to be beaten into thin sheets) and conductivity. According to her team, the first two attributes came by compromising on conductivity. It was eventually discovered that while making polymers (scientific term for plastic) moldable, their structures were trapped in rigid forms thereby preventing electric current to pass through them. The researchers relaxed the structure of the plastics by treating them with an acid after processing.

Circuits are made by first coating the polymer material onto a plastic base substrate using a process called spin-coating. This involves placing a drop or two of the coating polymer on the substrate which is then spun at a fast speed. The spinning ends up spreading the polymer film evenly on the substrate whose thickness is about 100 nanometers. This is followed by evaporating gold on the film using a mask which is placed on the polymer film and then put in a gold evaporating contraption. The pattern of the mask determined where the gold will be evaporated.

Plastic Logic's companion epaper as an accessory to your mobile phone allows reading documents, attachments, news articles on the go very easy

The materials

Material matters a lot when it comes to making plastic conductive. Cambridge University's Cavendish Professor of Physics Richard Friend is one of the pioneers of the study of organic polymers and the electronic properties of molecular semiconductors. "Polymer semiconductors are developed for their own specific application. For example colour of light emissions is important in OLEDs whereas speed of switching is important in FETs", he said.

Even at Princeton University, some researchers from the team make new materials, others characterise the structure of these materials and others incorporate these materials to understand their potential in certain applications. An example is polyaniline, a conducting polymer which changes color when different voltages are applied to it. Such a polymer can be used as a sensor which changes colour when it comes in contact with a reagent, for instance in purifying water.

Simpler manufacturing

Another advantage with plastic electronics lies with the simple manufacturing process involved. Chip makers have to invest in large fabrication plants where silicon-based electronic chips are mass produced under the most cleanest of environment using the most expensive of manufacturing processes. You require very high temperatures for processes such as etching, lithography and extremely clean room facilities. We are all familiar with those space-ship-type suits which engineers wear before entering a fab. Polymer based electronics, on the other hand do not need any of the above-mentioned meticulous processes for production. Polymers can be printed out using traditional inkjet printers.